Pranav Kondapalli

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Education

Georgia Institute of Technology — Atlanta, Georgia
Master of Science in Computer Science: Artificial Intelligence — *Expected December 2027*

Georgia Institute of Technology — Atlanta, Georgia
Bachelor of Science in Computer Science - GPA: 3.65/4.00 — *May 2026*

- **Coursework:** Machine Learning, Intro to AI, Automata & Complexity, Design & Analysis-Algorithms, Computer Org./Programming, Data Structures & Algorithms, Object-Oriented Programming, Human-Centered Computing/Design

Experience & Research

Software Engineer Intern — May 2025 - September 2025
Florida Department of Commerce — *Tallahassee, Florida*

- Automated SQL data retrieval for daily spreadsheets used in team meetings to track completed work, unresolved issues, and project status, reducing preparation time by **50–75%** and saving **30–45 minutes** per day
- Developed Oracle **SQL** queries retrieving claimant records for information requests, cutting ticket-processing time by **40%**

Undergraduate Researcher — August 2024 - December 2025
Georgia Institute of Technology — *Atlanta, Georgia*

- Led the finance research group in developing reinforcement learning **(DQN and DDPG)** portfolio-management models with **Python and TensorFlow**, incorporating indicators, experience replay, target networks, and cost-aware reward functions
- Improved simulated returns by **7%** vs. a market-long baseline with DQN and cut portfolio volatility **30%** using DDPG
- Independently evaluated **Ridge, gradient-boosted trees**, and a **PyTorch neural network** for QQQ options across **9 model fits** and **18** simulated accounts over **3** chronological folds, incorporating transaction fees, settlement, and cash constraints
- Recorded a **7.96%** average simulated return with Ridge Regression across 3 separate $10,000 accounts

Software Engineer Intern — May 2024 - September 2024
AMG Invest — *Houston, Texas*

- Developed and deployed investment dashboards **(Next.js/React)** for **100+** traded stocks, enabling investors to run trading algorithms, review results, and monitor performance; automated application updates and deployments via **CI/CD** pipelines
- Created custom interactive **D3.js** and **Highcharts.js** visualizations to showcase data from investor-provided endpoints, displaying stock performance, algorithm outputs, and trading results across dashboards and related application pages
- Integrated client-provided **REST API** endpoints for stock and algorithm data; reduced retrieval and loading time by **17.5%**

Software Engineer Intern — May 2023 - August 2023
AMG Invest — *Houston, Texas*

- Led the design and development of a full-stack **Next.js/React** application with an interactive UI for **15+** investors, enabling execution of trading algorithms, results analysis, and performance tracking; deployed the application on **AWS Amplify**
- Implemented authentication and user management with **NextAuth.js**, **Prisma**, and **MongoDB**; integrated **AWS Lambda**, **S3**, and **DynamoDB** to store and retrieve investor data for the application's trading workflows

Projects

Machine Learning for Poker Opponent Modeling | *Python, Node.js, LightGBM, Playwright, XGBoost, CatBoost* — GitHub | Paper

- Built a poker hand-history analysis pipeline with game-state tracking and **CFR/DCFR** policies to evaluate legal actions
- Predicted opponent actions from player stats and hands using supervised learning (**Gradient-boosted Trees, Extra Trees**)
- Trained **9 opponent models** on **4M+ decisions**, cutting prediction log loss **18.2%** vs. prior baseline on **23** simulated games
- Achieved **67.37%** postflop prediction accuracy with LightGBM and **65.73%** multiway prediction accuracy with CatBoost

Prompt Compression for Sentiment Classification | *PyTorch, DistilBERT, Hugging Face Transformers, scikit-learn* — GitHub | Paper

- Rebuilt a **DistilBERT (transformer)** sentiment-classification pipeline on **56k SST-2 examples**, comparing random, inverse document frequency (IDF), and learned prompt compression to shorten example text and classify movie reviews faster
- Cut input tokens **28.6%** and GPU pipeline time **48.8%** vs. full prompts at **90.9%** accuracy using cached IDF compression

StackHUD - Poker Analytics Platform | *TypeScript, Next.js, React, Prisma, Fastify, Node.js, PostgreSQL, Firebase* — Live Site

- Created a full stack poker analytics app for **1,000+ registered users**, generating **$2K–$3K/mo** in subscription revenue
- Built a resumable data-import **(ETL)** pipeline processing **400K+** hands and **5.5M+** actions into searchable game histories
- Reduced dashboards' response time by **89%**, from 6.77 to 0.75 seconds by optimizing PostgreSQL aggregation and joins

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Fastify, PyTorch, TensorFlow, scikit-learn, Transformers, Pandas, NumPy
Databases, Cloud & Tools: PostgreSQL, MongoDB, Oracle, Prisma, Firebase, AWS, DigitalOcean, Git, Docker, Playwright